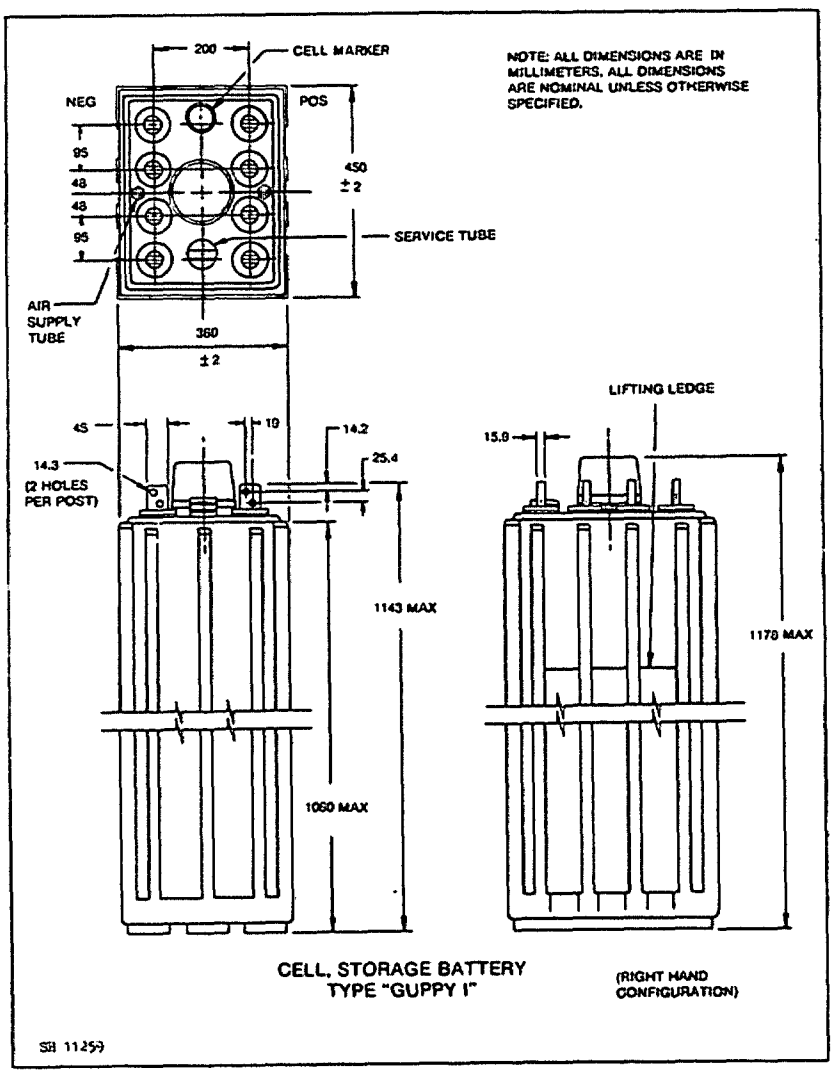
METRIC
DOD-B-24541/1A(SH)
24 November 1982
SUPERSEDING
DOD-B-24541/1(SH)
10 August 1976

MILITARY SPECIFICATION SHEET

BATTERY CELLS AND ELEMENTS, LEAD-ACID, MAIN STORAGE, SUBMARINE, TYPE GUPPY I (METRIC)

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery cells and elements described herein shall consist of this document and the latest issue of DOD-B-24541.



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FSC 6140

REQUIREMENTS:

Electrolyte specific gravity:	1.265 nominal.
Measurement of capacity:	Capacity, in ampere-hours (Ah), is based on an initial electrolyte temperature of 26.7 degrees Celsius specific gravity of 1.265, and shall be not less than as specified in table I.
Water cooling:	Not required.

TABLE I. Electrical characteristics.

Capacity Ah	Time hours	Discharge current A	Minimum battery voltage		Minimum cell voltage	
			126-cell			
			Average	Final	Average	Final
3390	1/2	6775	195	160	1.31	1.05
4340	1	4340	212	180	1.46	1.28
5500	3	1835	231	201	1.61	1.51
6100	5	1220	236	207	1.65	1.57
6270	6	1045	238	208	1.66	1.58
6850	10	685	241	212	1.69	1.63
7400	20	370	244	215	1.71	1.66
7740	36	215	246	217	1.72	1.68
7920	48	165	247	217	1.72	1.68

Mass per cell (filled and charged):	Maximum	- 470 kilograms (kg)
	Maximum average	- 465 kg
	Mininum	- 438 kg
Guaranteed service life: (equivalent cycle equals 5000 Ah of charge)	100 equivalent cycles or 48 months, whichever occurs first.	
Demonstration of rated capacity:	Within 30 equivalent cycles.	
Life test:	Accelerated float charge and ship float simulation regimes.	
Hydrogen emission (maximum): (measurements recorded as cubic centimeters/hour/cell/1,000 Ah at the 10-hour rate)	5,500 at 200 A 3,500 at 100 A 125 on open circuit	
Trickle discharge rate:	5A	

Preparing activity:
Navy - SH
(Project 6140-N561)

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